United International University
Department of Computer Science and Engineering

CSE 3421: Software Engineering Mid: Summer 2025

Total Marks: 30 Time: 1 hour and 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all five questions. The numbers on the right of the questions denote their marks.

1. (a) **Write GIT commands** for the following **instructions**: [CO2] (6)

   I **Download a repository** from **https://github.com/ShoeMart/ShopEase.git**.

   II Set **user name** as **ShopOps** and user email as **ops@shoemart.com**.

   III Create two new branches named **hotfix-payment** and **feature-discounts**.

   IV Move to the branch **hotfix-payment**, add file named **payment-fix.js**, then commit with message **"Fix payment rounding bug"**.

   V Add file named **coupons.md** to the branch **feature-discounts**, then commit with message **"Document coupon rules"**.

   VI **Delete coupons.md**, then commit with the message **"Remove coupons draft"**.

   VII **Delete the current commit** and go back to the previous commit (code: **f00d1123**).

   VIII Merge **all the branches into master** and **upload** the branch to GitHub.

2. (a) What is the **Requirements Specification** process? Mention **two models** that are used at this stage. (2)
   [CO1]

   (b) You are in charge of the online portal of a reputed university. The portal is facing the following issues: (4)
   Case 1: The university's learning management system (LMS) suddenly shuts down, with all course materials and student submissions encrypted. A note appears on the screen, demanding payment in bitcoin before midterm exam so that data can be recovered.
   Case 2: Some professors and students report that their portal accounts have been hijacked, with fake assignments or announcements posted. Logs show multiple login attempts from foreign IPs before access was gained.
   What kind of **cyber-attacks** might be responsible for this scenario? **Explain** with **proper reasoning**. [CO2]

3. (a) At a weekend launch event for Pathao Pay in Dhaka, a new hire is asked to compare two write-ups. One says, "Our platform splits into clearly bounded services. Requests are checked at the edge, authenticated with signed tokens, logged to a wallet ledger, and any temporary failures are queued on schedule." The other says, "Pay in seconds across citywide partners and grab attractive cashbacks on first payments." (2)
   Identify which type of **documentation** each write-up represents with **proper reasoning**. [CO1]

   (b) The Election Commission of Bangladesh is planning to introduce a new biometric voter verification system for the upcoming national elections. Since the system will be deployed across rural areas where internet connectivity is uncertain, different devices and software solutions need to be tested in small pilot projects first. Officials also require formal evaluations and approvals after each stage of development to ensure reliability before the nationwide rollout. (4)
   Which **software development model** would be most suitable for this project? **Explain** your reasoning clearly. [CO2]

4. (a) How does the **Daily Scrum** meeting differ from the **Sprint Planning** meeting in Scrum? Write any **two clear differences**. [CO2] (2)

   (b) Chaldal's SWE team works side by side with Shila from category management all day. She answers questions as they come up and tries each change herself, then gives a quick yes or no. Before adding the new delivery-fee rule, Rafi writes a small code check that fails so the team knows the target, and then adds minimal passing code. After that, Fatema cleans the code by improving names and breaking big functions into smaller ones while keeping the behavior the same. Each change is merged to the main line, an automatic run checks everything, and the team fixes any break before doing more work. (4)
   Which **XP practices** are reflected by the team's actions? **Briefly explain** with **proper reasoning**. [CO2]

5. (a) What kind of problem arises if **methods in multiple subclasses show a similar behavior**? How would you **refactor** such codes? [CO2] (2)

(b) **Refactor** the following code: [CO2] (4)

```
public class X {
    String x;
    int y;

    X(String x, int y) {
        this.x = x;
        this.y = y;
        System.out.println("Sport: " + x + " is played among " + y + " players");
        if (y > 10) System.out.println("Big team game");
        if (y > 5) System.out.println("Medium team game");
        if (y > 0) System.out.println("Small team game");
    }
}
```

```
public class Y {
    String z;
    int k;

    Y(String z, int k) {
        this.z = z;
        this.k = k;
        System.out.println("Tournament: " + z + " is created with " + k + " teams.");
    }

    void m() {
        int n = 1;
        for (int i = 1; i <= k; i++) {
            for (int j = 1; j <= k; j++) {
                if (i == j) n = n + 0;
                else n = n + 1;
            }
        }
        n = n/2;
        System.out.println("Total number of matches: " + n);
    }
}
```